

# WASP-135b

R-2619

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-135\_260724 · created 2026-08-28T23:18:04+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2461245.735 +/- 0.013 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.171 +/- 0.05                |
| Transit depth (Rp/Rs)^2                           | 2.91 +/- 1.7 [%]              |
| Orbital Inclination (inc)                         | 83.8 +/- 2.66                 |
| Airmass coefficient 1 (a1)                        | 0.332 +/- 0.016               |
| Airmass coefficient 2 (a2)                        | 1.07 +/- 0.39                 |
| Scatter in the residuals of the lightcurve fit is | 4.73 %                        |
| Best Comparison Star                              | #1 - [449, 308]               |
| Optimal Method                                    | PSF photometry                |
| Transit Duration (day)                            | 0.071 +/- 0.016               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                      |
|--------------------------------------|------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.171 $\pm$ 0.05 vs 0.139 (deviation 0.47 $\sigma$ ) |
| Duration fit vs published            | 0.071 d vs 0.0687 d (deviation 0.1 $\sigma$ )        |
| Mid-time O-C                         | 38.4 min                                             |
| Depth SNR                            | 2.5                                                  |
| Residual scatter                     | 4.73 %                                               |

### Light curve

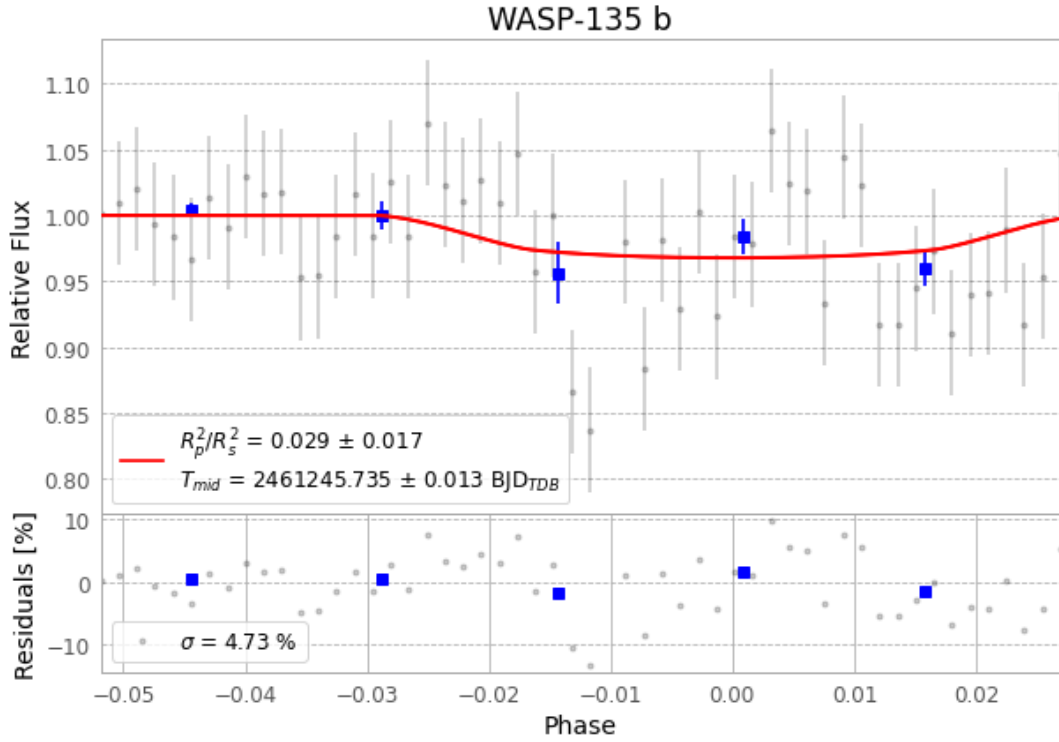


Figure 1 — FinalLightCurve\_WASP-135 b\_2026-07-23.png (SHA-256 ebc6d018af678ee8...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [227, 201], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 79189c10a07f68a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

55 FITS frames (36 MB), WASP-135260724034816.FITS → WASP-135260724063015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-260724.log (b55b1d11088f...), FinalLightCurve\_WASP-135 b\_2026-07-23.png (ebc6d018af67...), FinalLightCurve\_WASP-135 b\_2026-07-23.csv (c2a11a7289ea...)

## Compute resources

Wall time 133.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:18Z.